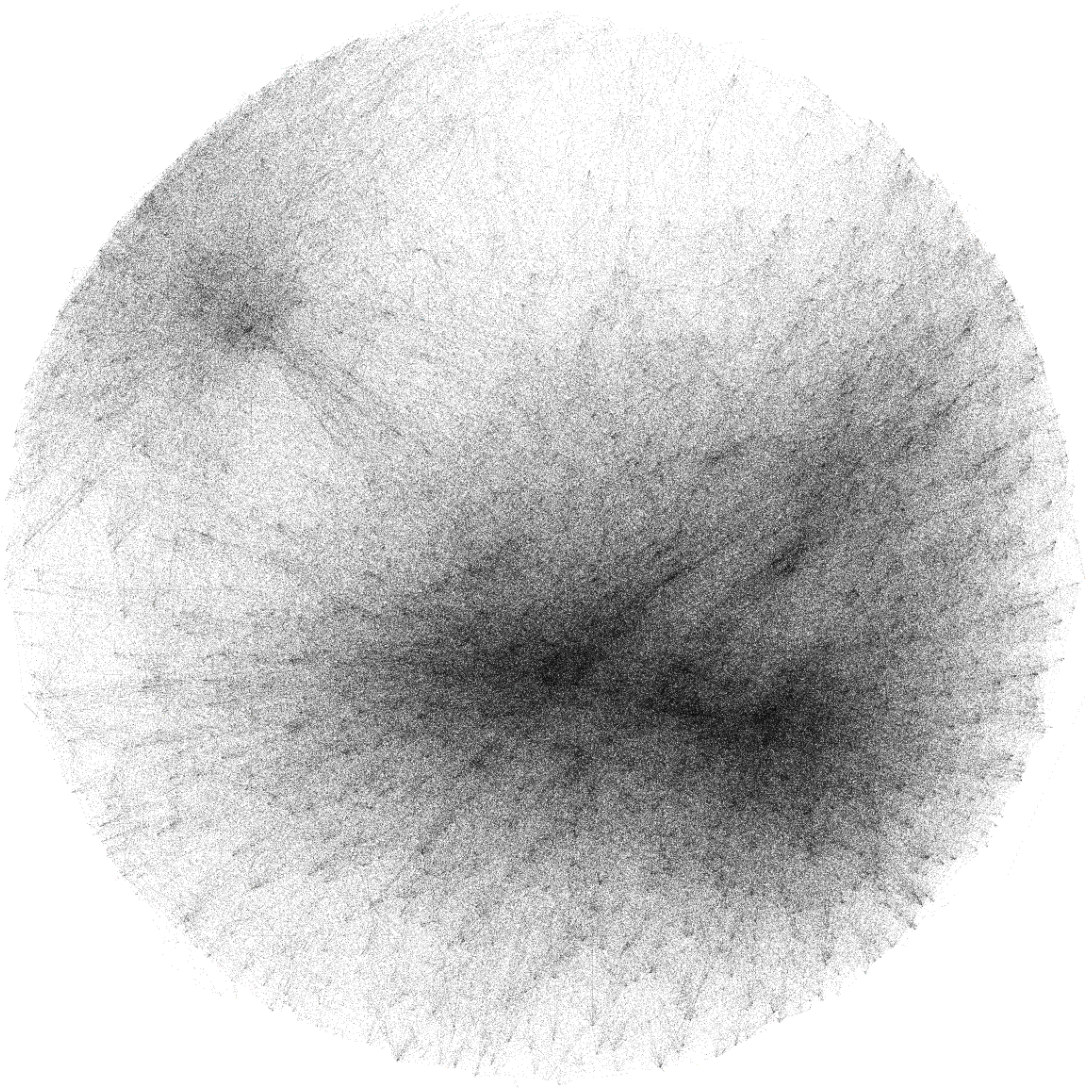
Assignment 2: Dynamics on Networks: Thresholds and Spreading in a Real-World Facebook Network

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Introduction

The ways in which information spreads through a social network hinges upon a combination of factors, including topology, initial condition, and ease of diffusion. In threshold-based diffusion models, like the Granovetter threshold model, cascades can occur when low-threshold/vulnerable nodes adopt information, enabling propagation through the network (Granovetter, 1978; Watts, 2002). When such vulnerable nodes cluster together, a single adoption can cascade and push the system past a tipping point, producing large-scale adoption.

This analysis examines the dynamics of the Granovetter threshold model on a real-world social network, focusing on how network topology influences information diffusion under various seeding strategies. The real-world network in this analysis constitutes a Bowdoin College Facebook network. Bowdoin College is a liberal arts institution in Maine, USA, with a student population of approximately two thousand. The dataset represents an undirected, unweighted social network, where nodes correspond to Facebook users affiliated with Bowdoin College and edges represent mutual friendships (Rossi and Ahmed, 2015).

Methods

To examine the influence of network topology on information diffusion, three seeding strategies were implemented: (1) random node seeding, serving as a control to assess the baseline spread of information across the Facebook network; (2) highest-degree node seeding, used to evaluate the effect of node popularity on information propagation; and (3) highest-betweenness node seeding, used to investigate how nodes functioning as brokers along shortest paths between other nodes influence the spread of information. In addition to seeding strategies, this analysis compared two threshold modes. First, a fixed threshold attribution assigned all nodes with the same adoption threshold. Second, a β threshold mode attributed adoptions thresholds from a distribution concentrated around a mean:

$$E[X] = \frac{\alpha}{\alpha+\beta}$$

With the α and β parameters, the distribution could be skewed toward 0 or 1. Both threshold modes assigned thresholds as a fraction of adopted neighbors. This is expressed in the update rule:

$$x_i(t+1) = \begin{cases} 1, & \text{if } \dfrac{\sum_{j\in\mathcal{N}(i)} x_j(t)}{k_i} \geq \phi_i, \\ x_i(t), & \text{otherwise.} \end{cases}$$

Here, $x_i(t) \in \{0, 1\}$ represents the adoption state of node i at time t, k_i is the degree and $\mathcal{N}(i)$ is the neighborhood of i, $\phi_i \in [0, 1]$ represents the adoption threshold of i, and lastly, adoption in this model occurs monotonically: $0 \to 1$, without recovery (Lees, 2025).

Table 1: Summary of network and experiment parameters.

Parameter	**Value**	**Explanation**
N	2250	Number of nodes in the network
L	84386	Number of edges in the network
B	22	Number of initial adopters ($\sim$1% of the network)
Runs per setting	50	Number of simulations per threshold setting
Threshold modes	Fixed and β distribution	Modes for threshold attribution

The number of initial adopters was set at 22 so it is small enough that cascades are not trivially guaranteed and large enough to produce reliably measurable outcomes across replicate runs. Runs per setting was set at 50 to strike a pragmatic balance between estimating the true (stochastic) behavior of the system and limiting computational cost.

Results

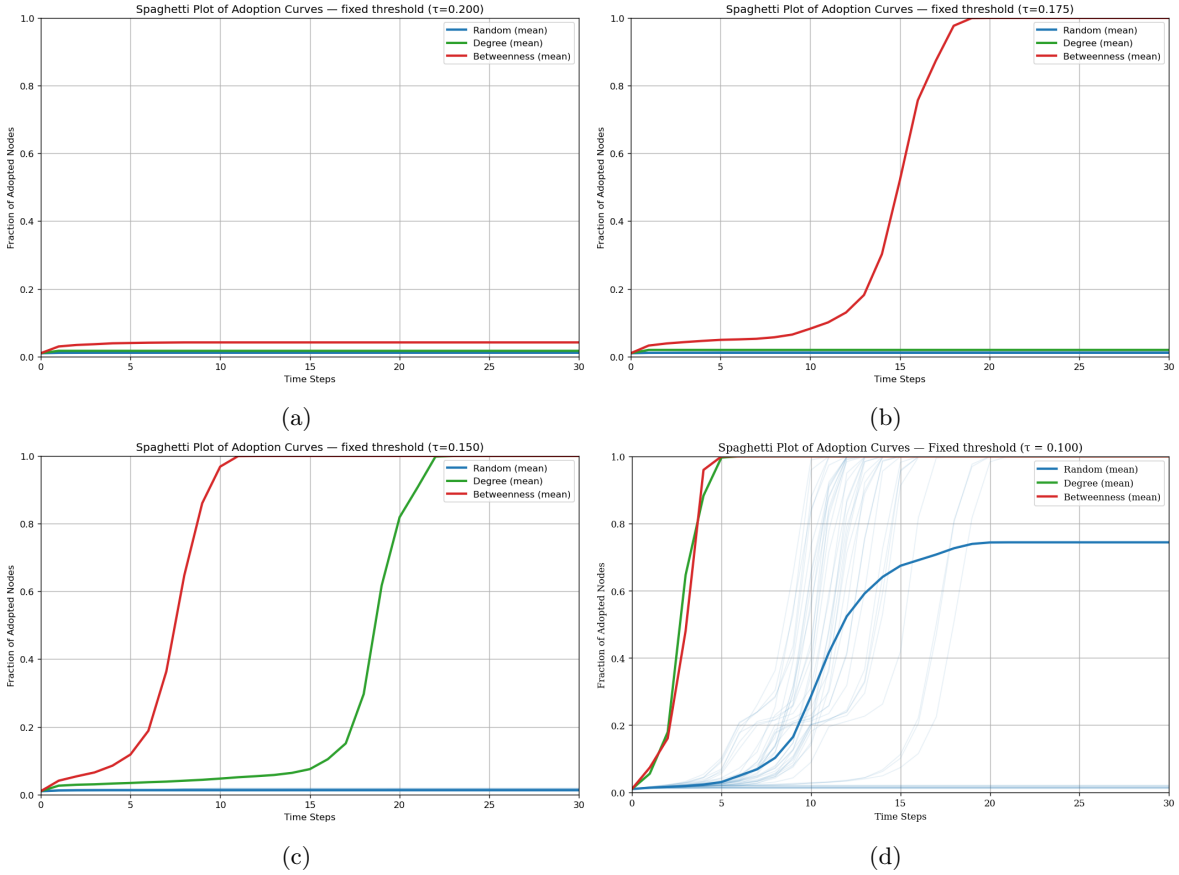


Figure 1: Fixed Threshold Adoption Curves Over Time. From (a) ($\tau = 0.200$) to (d) ($\tau = 0.100$), as the adoption threshold decreases, seeding strategies sequentially meet the cascade condition in order of betweenness → degree → random seeding strategies. In (d), random seeding produces variation across simulation runs (N=50) because each run starts with a different set of initial nodes, whereas centrality-based strategies yield consistent results since the same nodes are always seeded, leading to predictable paths of information spread.

Simulations of the Granovetter threshold model on the Bowdoin Facebook network reveal distinct dynamics depending on both the seeding strategy and the threshold mode. In the fixed threshold mode, a clear hierarchy arises in cascading and timing: at $\tau = 0.175$, only simulations seeded based on betweenness centrality met the cascade condition (fig. 1b). At $\tau = 0.150$, both centrality-driven simulations met the cascade condition with the betweenness strategy doing so at $t = 11$ and the degree strategy at $t = 22$ (fig. 1c). Random seeding lagged behind, only starting to meet the cascade condition in a portion of simulation runs at $\tau = 0.100$ (fig. 1d). When adoption thresholds were drawn from a beta distribution, however, the distinction between betweenness- and degree-based seeding changed (fig. 3). Degree-based seeding lead to more cascades (fig. 3b). Additionally, cascade conditions were met for the β simulations at higher mean thresholds than their fixed counterparts. Namely, at $\tau = 0.200$, none of the simulations met the cascade condition, whereas at a β distribution with $\alpha = 2$ and $\beta = 8$, a mean threshold of 0.200 arises and all simulations met the cascade condition (figs. 1a, 2d & 3d)

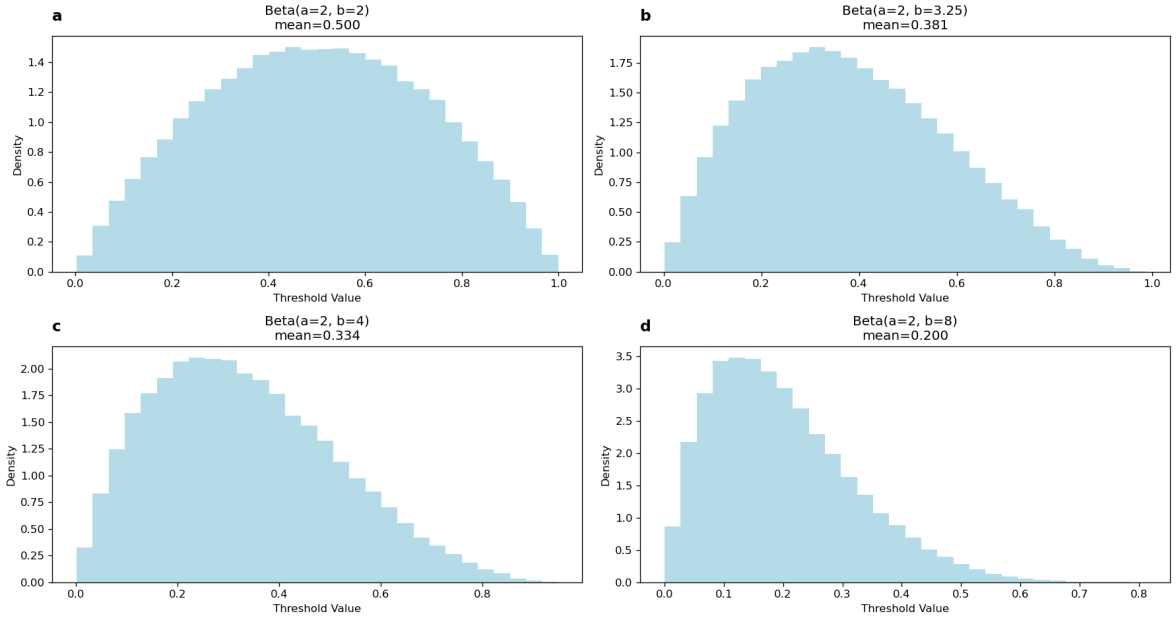


Figure 2: β distributions of adoption threshold values for various α and β values (N=100000)

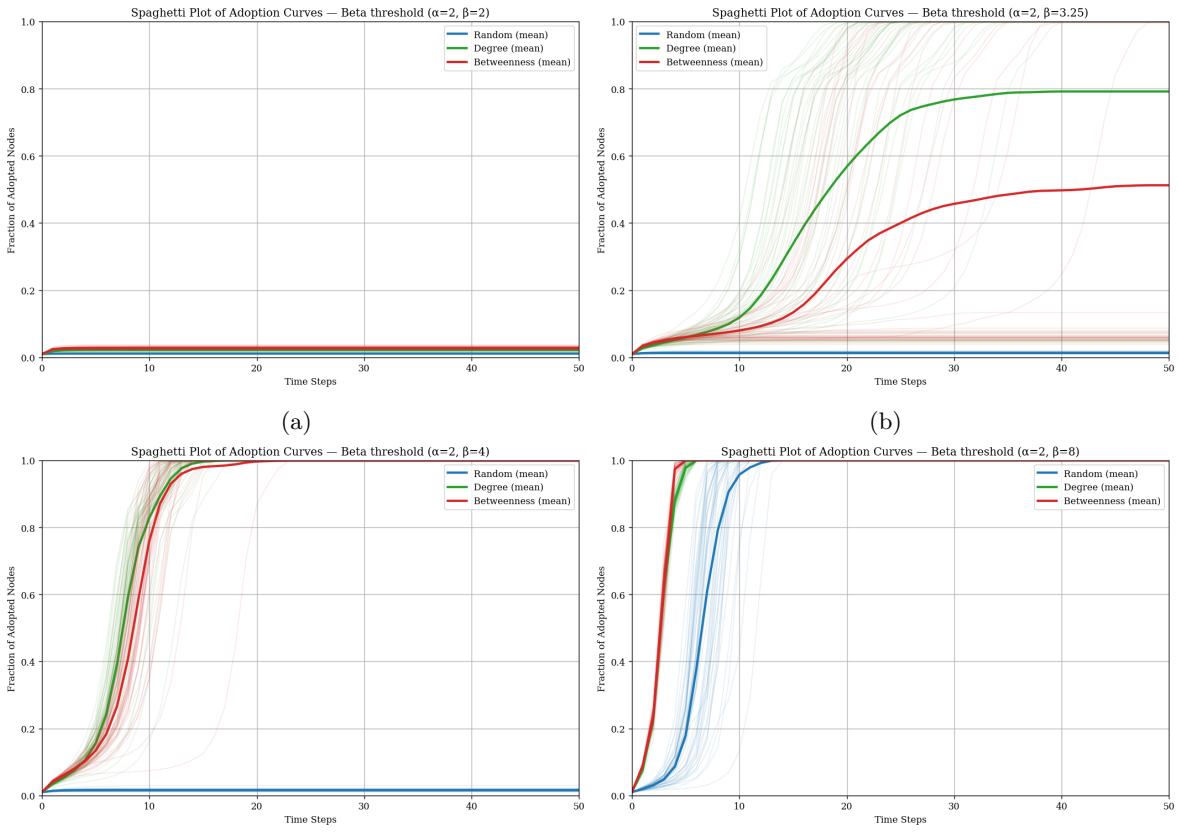


Figure 3: β distribution adoption curves over time (N=50). From (a) to (d), as the ratio of α to β decreases, the fraction of adopted nodes for all seeding strategies increases (due to an increasingly skewed β distribution toward a lower mean threshold). The centrality-driven seeding strategies experienced full adoption with higher α to β ratios. Simulations in this figure do not draw from the β distribution in figure 2 due to constraints in computational power.

Discussion

This analysis demonstrates that information spread in the Bowdoin Facebook Network is governed by where it begins (seeding strategies), how difficult nodes are to activate (adoption thresholds), and the underlying network topology. In the Bowdoin Facebook network, these factors created differences in whether, when, and how fast cascades occured.

Across the fixed-threshold simulations, clear differences in cascade sensitivity and timing arose. These differences reflect a positional advantage of nodes that lie on shortest paths between nodes (fig. 1a & 1b). Activating them allows information to bypass local bottlenecks and penetrate multiple network regions before local clusters have fully reacted. High-degree seeds, though well connected, primarily accelerate adoption in their immediate neighborhoods, and only trigger system-wide cascades when thresholds are sufficiently low or when those neighborhoods include many vulnerable nodes. Lastly, lacking any structural advantage, random seeding does not generate cascading conditions until thresholds are so low that diffusion is driven more by ease of activation than by the initial seed placement (fig. 1d). However, in general, fixed-threshold simulations are not representative of real-world social networks because people in social networks exhibit heterogeneity across their adoption thresholds—some people are more susceptible to adopting a piece of information than others.

Information diffusing through this network may take the form of a rumor, a meme, a news item, or any other shareable social media content. One is less likely to spread a negative rumor about a friend, but more likely to share a positive news item about a politician they support. For example, recently, Zohran Mamdani, a Bowdoin alum, won the New York City mayoral elections. The news of his electoral victory was shared on the Bowdoin College Facebook page. This page likely forms a hub in the Bowdoin College network. Alumni who like Mamdani's politics likely had a low threshold for sharing the post of his win, while those who do not like his politics likely had a much higher threshold, potentially even a threshold > 1. Although the data set used in this analysis does not include any information for identifying Facebook accounts, and this specific example might yield a polarized threshold distribution, these kinds of explanations are a real-world approach to using centrality-driven seeding strategies; the Bowdoin College Facebook page is likely the top node in terms of both betweenness and degree centrality, connecting clusters of class years, or even friendship groups to each other.

With this real-world scenario in mind, the introduction of heterogeneous thresholds via the β distribution illuminate the role of individual node susceptibility to adoption. Namely, heterogeneous thresholds increase the likelihood of cascades by creating pockets of vulnerable nodes (Watts, 2002). So, when adoption thresholds are variable rather than uniform, cascades emerge more frequently than in the fixed-threshold regime. This occurs because even a small proportion of vulnerable nodes can act as instigators. Once these instigators propagate forward, mini cascades perpetuate into the wider network. This could also explain the shift in cascade dominance from betweennes to degree seeding between the fixed and β threshold modes; threshold heterogeneity creates vulnerable nodes everywhere, and high-degree nodes are best connected to them, making global bridging (high betweenness) less essential. Regardless, both centrality-driven strategies gain access to pockets of vulnerable nodes scattered throughout the network, leading to similar timing of cascades.

Altogether, this analysis shows that seeding strategy can compensate for higher mean thresholds, that threshold heterogeneity can compensate for poor seed placement, and that dense, well-connected regions can compensate for weak initial conditions.

References

Granovetter, M. (1978). Threshold models of collective behavior. American Journal of Sociology, 83(6), 1420–1443

Lees, M. (2025, October 23). Week 3 — Lecture 2: Dynamics on networks – Opinions and robustness [Lecture slides]. Model Based Decisions, University of Amsterdam

Rossi, R. A., & Ahmed, N. K. (2015). The Network Data Repository with interactive graph analytics and visualization. In Proceedings of the AAAI Conference on Artificial Intelligence. Retrieved from https://networkrepository.com

Watts, D. J. (2002). A simple model of global cascades on random networks. Proceedings of the National Academy of Sciences, 99(9), 5766–5771. https://doi.org/10.1073/pnas.082090499